

Chapter 19

Chemical Thermodynamics

Review

ทิศทางของปฏิกิริยา

- **Spontaneous process:** ^{exothermic ปล่อย endo ปล่อย} one that proceeds on its own without any outside assistance, processes that are spontaneous in one direction are nonspontaneous in the opposite direction.
- **Reversible process:** can restore the system to its original condition with no net change to either the system or its surroundings. ^{equilibrium : interconverting at same rate}
- **Irreversible process:** one that cannot simply be reversed to restore the system and its surroundings to their original states. ^{heat flow}
 - 1) In the isothermal reversible process, the system do the greatest work to the surrounding;
 - 2) The reversible process is an ideal process, all real processes are irreversible; ^{infinitely slow change in real world}
 - 3) Any spontaneous process is irreversible;
 - 4) Reversible process is not the reversible reaction;

- **Enthalpy ($H=E+pV$):** the thermodynamic quantity equivalent to the total heat content of a system. It is equal to the internal energy of the system plus the product of pressure and volume.

- **Enthalpy change:**

- 1) When there is only the volume work, the enthalpy change equals the sum of the change of internal energy and the volume work under constant pressure ($\Delta H = \Delta E + p\Delta V$).
- 2) The enthalpy change is the heat absorbed by a system during a constant pressure process ($\Delta H = q_p$).

- **Entropy (S):** a measure of the randomness or disorder in a system. $J/mol \cdot K$

- **Entropy change (ΔS):**

- 1) Depend only on the initial and final states of the system and not on the path taken from one state to the other.

$$\Delta S = S_{final} - S_{initial} = \frac{q_{rev}}{T} \text{ (constant T)}$$

^{reversible}

$$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T}$$

- 2) The entropy change in a chemical reaction equals the sum of the entropies of the products minus the sum of the entropies of the reactants:

$$\Delta S^0 = \sum n S^0(\text{products}) - \sum m S^0(\text{reactants})$$

- **The first law of thermodynamics:** energy cannot be created or destroyed, but can be converted from one to another.

$$\Delta E = q + w$$

- **The second law of thermodynamics:** any irreversible process results in an increase in the entropy of the universe, whereas any reversible process results in no change in the entropy of the universe.

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 0 \text{ (reversible process)}$$

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0 \text{ (irreversible process)} \rightarrow \text{Spontaneous}$$

- **Boltzmann equation:** entropy is a measure of how many microstates are associated with a particular macroscopic state.

$$S = k \ln W \quad \leftarrow \# \text{ microstates}$$

$$\Delta S = k \ln W_{final} - k \ln W_{initial} = k \ln \frac{W_{final}}{W_{initial}}$$

^{$1.38 \times 10^{-23} J/K$}

k: Boltzmann constant = 1.38×10^{-23} J/K;

W: number of microstates of a system;

ΔS : entropy increases with the number of microstates of the system;

- **Microstate:**

- 1) a single possible arrangement of the positions and kinetic energies of the molecules when the molecules are in a specific thermodynamic state;
- 2) a particular microscopic arrangement of the atoms or molecules of the system that corresponds to the given state of the system;

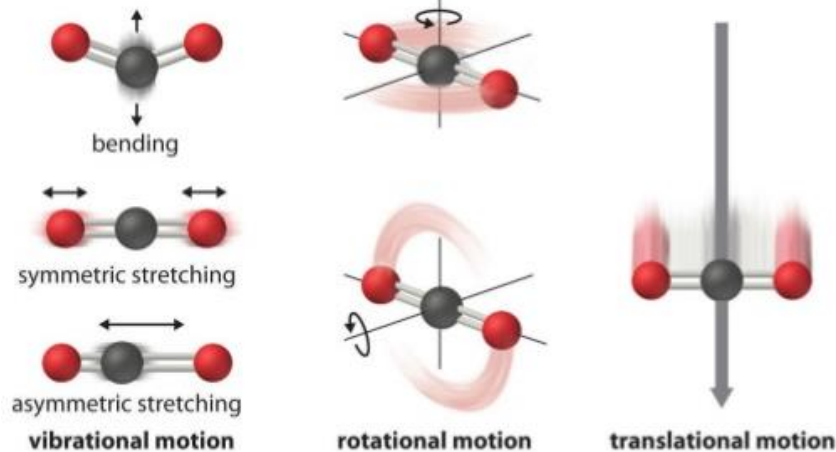
- The number of microstates (the entropy) tend to increase with increases in:

- 1) Temperature; speed distribution ↑
- 2) Volume; $\text{volume} \uparrow$
- 3) The number of independently moving molecules; ice < water < steam

- **Molecular motion:**

- 1) **Translational motion:** movement of the entire molecule from one place to another; translation
- 2) **Vibrational motion:** the atoms in the molecule move periodically toward and away from one another; vibration
- 3) **Rotational motion:** the molecule spins about an axis. rotation

Note: More atoms means more microstates (more possible molecular motions).



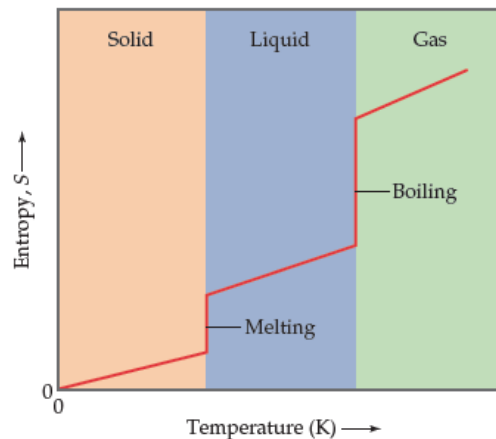
- **The third law of thermodynamics:** the entropy of a pure, perfect crystalline substance at absolute zero is zero: $S(0 \text{ K}) = 0$

perfectly ordered, 1 possible microstate

- **Standard molar entropy (S_m°):** molar entropies for substances in their standard states ($\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

$$\Delta S^\circ = \sum n S^\circ_{\text{products}} - \sum m S^\circ_{\text{reactants}}$$

- 1) Standard molar entropy of gas is greater than that of liquid and solid;
- 2) It increases with the increasing molar mass for similar substances;
- 3) It increases with an increasing number of atoms in the formula of substances; $\# \text{ moles}$
- 4) It increases with the structural complexity of substances which have the same molar mass;
- 5) It increases with the higher temperature for the same substances;
- 6) The influence of pressure on solid and liquid is less, while for gas is great.



$$\Delta S_{\text{sur}} = \frac{-q_{\text{sys}}}{T} = \frac{-\Delta H_{\text{sys}}}{T} \quad (\text{Const } P)$$

- **Gibbs free energy (G):** $G = H - TS$ (state function)
- **Gibbs-Helmholtz Equation: for an isothermal process**

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$-\frac{\Delta G}{T} = -\frac{\Delta H}{T} + \Delta S$$

$$-\frac{\Delta G}{T} = \Delta S_{\text{surr}} + \Delta S_{\text{sys}} = \Delta S_{\text{univ}}$$

$$\Delta G = -T\Delta S_{\text{univ}}$$

$$(at\ constant\ P,\ T: \Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T} = -\frac{q_{\text{sys}}}{T} = -\frac{\Delta H_{\text{sys}}}{T}) \quad \text{Free energy indicates equilibrium}$$

- 1) If $\Delta S_{\text{univ}} > 0$ ($\Delta G_{T,p} < 0$), the reaction is spontaneous in the forward direction;
 - 2) If $\Delta S_{\text{univ}} = 0$ ($\Delta G_{T,p} = 0$), the reaction is at equilibrium;
 - 3) If $\Delta S_{\text{univ}} < 0$ ($\Delta G_{T,p} > 0$), the reaction is spontaneous in the reverse direction;
- **Standard Gibbs free energy of formation (ΔG_f°):** the change of Gibbs free energy that accompanies the formation of 1 mole of a substance in its standard state from its constituent elements in their standard states.

State of Matter	Standard State
Solid	Pure solid
Liquid	Pure liquid
Gas	1 atm pressure
Solution	1 M concentration
Element	$\Delta G_f^\circ = 0$ for element in standard state

- **Standard free-energy change for chemical processes:**

$$\Delta G^\circ = \sum n\Delta G_f^\circ(\text{products}) - \sum m\Delta G_f^\circ(\text{reactants})$$

- **Free-energy change under nonstandard condition:**

$$\Delta G = \Delta G^\circ + RT \ln Q$$

R: ideal-gas constant;

T: absolute temperature;

Q: the reaction quotient for the reaction mixture of interest;

Std. conditions $\rightarrow Q = 1$

Equilibrium $\rightarrow Q = K$ ($\Delta G = 0$)

Gases conc $\rightarrow Q_p, K_p$

- **Relationship between ΔG° and K: at equilibrium, $\Delta G = 0$, $Q = K$**

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

$$\ln K = \frac{\Delta G^\circ}{-RT}$$

$$K = e^{-\Delta G^\circ/RT}$$